

A
MAJOR PROJECT REPORT
ON
IOT BASED AUTOMATED GRAIN DRYING AND RAIN
PROTECTION SYSTEM USING TELEGRAM ALERTS

Submitted in partial fulfilment of the requirement
for the award of the degree
BACHELOR OF TECHNOLOGY

In
INFORMATION TECHNOLOGY

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Department of Information Technology
VIGNANA BHARATHI INSTITUTE OF
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Academic Year 2025-26



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CERTIFICATE

This is to certify that the project entitled “**IOT BASED AUTOMATED GRAIN DRYING AND RAIN PROTECTION USING TELEGRAM ALERTS**” is being submitted by M. KEERTHANA (22P61A1271), P.RAGHURAM (22P61A1288), P.RUCHITH (22P61A1289), T. NAGASRI (22P61A12B5) in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Information Technology is a record of Bonafide work carried out by them under my guidance and supervision during the academic year 2025- 2026.

The results embodied in this project report have not been submitted to any other University for the award of any degree or diploma.

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DECLARATION

We, **M. KEERTHANA** bearing hall ticket number **22P61A1271** , **P. RAGHURAM** bearing hall ticket number **22P61A1288**, **P.RUCHITH** bearing hall ticket number **22P61A1289** and **T. NAGASRI** bearing hall ticket number **22P61A12B5**, hereby declare that the project report entitled “**IOT BASED AUTOMATED GRAIN DRYING AND RAIN PROTECTION USING TELEGRAM ALERTS** ” under the guidance of **Mrs. N.INDIRA PRIYADARSINI**, Department of Information Technology, VBIT, Hyderabad, is submitted in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology in Information Technology.

This is a record of Bonafide work carried out by us and the results embodied in this project have not been reproduced or copied from any source. The results embodied in this project report have not been submitted to any other university or institute for the award of any other degree or diploma.

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ACKNOWLEDGEMENT

First and foremost, we wish to express our gratitude towards the institution “Vignana Bharathi Institute of Technology” for fulfilling the most cherished goal of our life to do Bachelor of Technology.

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We also express our sincere thanks to Mrs. V. Ambica, Project Coordinator for her encouragement and support throughout the project.

We are deeply grateful to Dr. V. Sridhar Reddy, Head of Department, Department of Information Technology for granting us the opportunity to conduct this project.

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COURSE OUTCOMES

Course: Major Project

Academic Year: 2025 - 2026

Class: IV B. Tech II Semester

Section: IT-B

Course Outcomes

After completing the Project, the student will be able to:

Code	Course Outcomes	Taxonomy
C424.1	Identify and clearly define the problem of grain damage due to rain, excess moisture, and fire risk, and prepare the project abstract accordingly.	Remember
C424.2	Analyze the existing agricultural practices and propose an IoT-based automated solution for grain drying and rain protection.	Analyze
C424.3	Apply modern tools such as sensors, microcontrollers, and communication platforms to design the system based on specified requirements.	Apply
C424.4	Develop and test an automated system that controls the shield roof, monitors grain moisture, detects fire, and sends Telegram alerts to ensure system reliability.	Create
C424.5	Prepare comprehensive project documentation including architecture, implementation, results, and conclusions as per institutional guidelines.	Create

PROGRAM OUTCOMES (POs)

1. **Engineering knowledge:** This project applies knowledge of electronics, sensors, microcontrollers, and basic engineering concepts to design an automated system for grain drying, rain protection, and fire detection. It uses engineering fundamentals to solve real agricultural problems.
2. **Problem analysis:** The project identifies problems such as crop damage due to rain, excess moisture in grains, and fire risk. These problems are analyzed using sensor data and environmental conditions to take correct automated actions.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the
4. **Conduct investigations of complex problems:** The project involves testing sensor readings, analyzing moisture and rain levels, and verifying system responses.

5. **Modern tool usage:** Modern tools such as IoT platforms, sensors, microcontrollers, and Telegram alert services are used. These tools help in real-time monitoring, automation, and remote communication.
6. **The engineer and society:** This project addresses social and agricultural challenges by helping farmers reduce crop loss, save labor, and improve productivity. It supports safer and smarter farming practices.
7. **Environment and sustainability:** By preventing grain spoilage and reducing wastage, the project promotes sustainable agricultural practices. Efficient drying and protection reduce the need for repeated farming and resource loss.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** The project is developed through teamwork, where members contribute to hardware setup, coding, testing, and documentation, demonstrating effective collaboration.
10. **Communication:** The project requires clear communication through reports, diagrams, and presentations. Telegram alerts also act as a communication medium between the system and farmers.
11. **Project management and finance:** The project is completed within a limited budget and time. Proper planning of components, cost management, and task allocation are involved in successful implementation.
12. **Life-long learning:** The project encourages learning new technologies such as IoT, sensor integration, and automation systems, motivating continuous learning to adapt to future technological advancements.

PROGRAM SPECIFIC OUTCOMES (PSO's)

PSO1	Apply knowledge of sensors, microcontrollers, and embedded hardware to design and simulate an IoT-based system for automated grain drying, rain protection, and fire detection.
PSO2	Analyze basic networking and communication concepts to enable real- time data transmission and Telegram-based alert notifications for remote crop monitoring.
PSO3	Design and manage simple data handling mechanisms for storing and processing sensor readings and system status to support efficient decision-making.
PSO4	Apply programming logic and control algorithms to develop an automated solution that responds to environmental conditions such as rain, moisture levels, and fire detection.



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Institute of Technology

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Department of Information Technology

COs Mapping with PO/PSO

Project Title: IOT BASED AUTOMATED GRAIN DRYING AND RAIN PROTECTION USING TELEGRAM ALERTS

Name of the Supervisor: Mrs. N. Indira Priyadarsini

Batch Details:

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04	22P61A12B5	T. Nagasri	

CO-PO Mapping for Major Project:

High -3 Medium -2 Low-1

PO/ CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O 1	PS O2	PS O3	PS O4
C414.1	3	3	3	3	3	3	2	2	3	3	2	2	3	2	2	3
C414.2	3	3	3	3	3	3	2	2	3	3	2	2	3	2	2	3
C414.3	2	3	3	3	3	3	2	2	3	3	2	2	3	2	2	3
C414.4	2	3	3	3	3	3	2	2	3	3	2	2	3	2	2	3
C414.5	2	3	3	3	3	3	2	2	3	3	2	2	3	2	2	3
AVG	2.4	3.0	3.0	3.0	3.0	3.0	2.0	2.0	3.0	3.0	2.0	2.0	3.0	2.0	2.0	3.0

CO-PO mapping Justification

Course: Major Project

Academic Year: 2026

Class: IV B. Tech II Semester

Section: IT-B

PO1	Students are able to apply knowledge of sensors such as rain, moisture, and fire sensors along with microcontrollers and motors to automate grain drying and rain protection.
PO2	Students are able to analyze real-world agricultural problems like crop damage due to rainfall, excess moisture in grains, and fire hazards using sensor data.
PO3	Students are able to design the complete system architecture including sensor integration, controller logic, shield roof mechanism, and Telegram alert system.
PO4	Students are able to investigate system performance by testing rain detection, moisture thresholds, fire sensing, and actuator responses.
PO5	Students are able to use modern tools such as IoT platforms, microcontrollers, programming environments, and Telegram bot services for automation and monitoring.
PO6	Students are willing to apply engineering solutions to support farmers by reducing crop losses, labor dependency, and improving agricultural productivity.
PO7	Students are willing to promote sustainable agricultural practices by minimizing grain spoilage, reducing wastage, and improving drying efficiency.
PO8	Students are willing to follow ethical responsibilities by ensuring safe system operation and responsible usage of alert and monitoring systems.
PO9	Students are able to work effectively as individuals and as a team during hardware setup, coding, testing, and project documentation.
PO10	Students are able to communicate effectively through project reports, diagrams, presentations, and Telegram alert notifications.
PO11	Students are able to apply basic project management skills including task planning, time management, component selection, and cost control.
PO12	Students are willing to engage in lifelong learning by adopting new IoT technologies, automation concepts, and real-time communication systems.

PSO1	Students are able to apply knowledge of sensors, embedded hardware, and IoT automation to develop an automated grain drying and rain protection system.
PSO2	Students are able to analyze networking and communication concepts to enable real-time data transmission and Telegram-based alert notifications.
PSO3	Students are able to manage sensor data, system status information, and alert notifications to support effective decision-making.
PSO4	Students are able to implement programming logic and control algorithms to respond automatically to rain, moisture levels, and fire detection.

Supervisor Signature

ABSTRACT

Agriculture faces many challenges due to unpredictable weather conditions and inefficient crop management practices. During the harvesting and post-harvesting stages, grains are usually dried in open fields under sunlight to reduce their moisture content before storage. However, sudden rainfall during this drying process can significantly increase the moisture level of grains, leading to fungal growth, spoilage, and reduction in grain quality. These conditions may result in considerable financial losses for farmers. In addition to rainfall, other issues such as improper irrigation, animal intrusion, and accidental fire hazards can also threaten crop safety and agricultural productivity.

To overcome these challenges, this project proposes an IoT-based automated grain drying and rain protection system with Telegram alerts. The system uses a combination of sensors and smart automation techniques to continuously monitor environmental conditions in agricultural fields. Sensors such as rain sensors, soil moisture sensors, temperature and humidity sensors, motion sensors, and fire sensors are used to collect real-time data related to the field environment. The collected data is processed by a microcontroller such as ESP32 or Arduino, which acts as the central control unit of the system. Based on the sensor readings, the controller automatically performs the necessary actions to protect crops and improve farming efficiency.

When rainfall is detected, the system automatically activates a motorized protective shield to cover the grains and prevent them from getting wet. When high humidity levels are detected, a grain drying fan is activated to reduce moisture and maintain optimal drying conditions. Similarly, the irrigation system is automatically controlled based on soil moisture levels to ensure efficient water usage and prevent over-irrigation. In addition to these features, PIR motion sensors detect the movement of animals near the crop area and trigger alerts to warn farmers about possible crop damage. A fire detection sensor is also included to identify fire hazards at an early stage and activate a buzzer alarm for immediate attention.

A key feature of the proposed system is the integration of the Telegram messaging platform for real-time notifications. Whenever an important event occurs, such as rain detection, activation of the grain dryer, animal intrusion, or fire detection, the system immediately sends alert messages to the farmer's mobile phone via Telegram. This enables farmers to remotely monitor their fields and stay informed about the condition of their crops even when they are not physically present at the farm.

The proposed system helps reduce manual effort, improves crop protection, and increases agricultural productivity through automation and smart monitoring. By integrating multiple agricultural functions such as grain drying, rain protection, irrigation control, and crop safety into a single IoT-based platform, the system provides a cost-effective and farmer-friendly solution for modern agriculture. This technology supports the concept of smart farming, where sensors, automation, and communication technologies work together to enhance crop management and reduce losses caused by environmental factors.

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CHAPTER 1
INTRODUCTION

1. INTRODUCTION

Agriculture plays an important role in the economy, and grain drying is a critical process after harvesting to reduce moisture content. Sudden or heavy rainfall during drying can damage grains, reduce quality, and cause financial loss to farmers. Traditional methods require constant manual supervision and are not effective during unexpected weather changes. To overcome these problems, this project proposes an IoT-based automated grain drying and rain protection system using an ESP32 microcontroller. The system uses rain and moisture sensors to detect environmental conditions and automatically activates a protective shield and drying fan when required. Real-time Telegram alerts keep farmers informed about the system status, reducing manual effort and improving grain protection efficiency.

The project titled “**IoT Based Automated Grain Drying and Rain Protection Using Telegram Alerts**” is designed to protect grains from rain and reduce moisture automatically. An ESP32 microcontroller is used as the main controller. A rain sensor detects rainfall and activates a protective shield to cover the grains, especially during heavy rain. Soil moisture sensors measure the moisture level of the grains, and when moisture is high, a drying fan is turned ON. The system also sends Telegram alerts to the farmer so that they can know the condition of the grains from anywhere. An LCD display shows the current system status. This system is simple, cost-effective, reduces manual work, and helps farmers protect their crops efficiently.

1.1 OVERVIEW OF EXISTING SYSTEM

In the existing system, farmers usually sell their produce through middlemen or local markets. These intermediaries control pricing, which often results in low income for farmers. There is a lack of transparency in market rates and demand, and most transactions are done manually. Farmers do not have direct access to buyers, which limits their profit and growth. Overall, the current system is inefficient, time-consuming, and unfavorable to farmers.

- The system includes an **automated shield/roof mechanism** to protect crops or grains from rain.
- **Rain sensors** are used to detect rainfall automatically.
- When rain is detected, the **shield opens automatically** to prevent rain damage.
- The system also supports **grain drying**, where grains are exposed to sunlight when there is no rain.
- The shield closes during dry conditions to **allow proper drying of grains**.
- **Fire detection sensors** are used to detect fire or high temperature in crop storage or field areas.
- When fire is detected, the system gives an **alert or warning signal**.
- A **controller unit** manages rain detection, shield movement, grain drying operation, and fire detection.

1.2 OVERVIEW OF PROPOSED SYSTEM

In the proposed system, an IoT based automated grain drying and rain protection system is developed to protect crops from rain and excess moisture. An ESP32 microcontroller is used to control the entire system. A rain sensor detects rainfall. When rain occurs and the soil moisture level becomes greater than 50%, a protective shield (roof) automatically opens to cover and protect the crops. This prevents rainwater from directly falling on the grains. If the grains contain excess moisture, a drying fan is turned ON to reduce the moisture content. When the rain stops and the moisture level comes below 50%, the shield closes automatically and the drying process continues. The system sends Telegram alerts to the farmer so they can know the condition of the crops from anywhere. An LCD display shows the system status and sensor values. This system works automatically and reduces manual work.

- Automatic rain detection
- Soil moisture monitoring
- Automatic shield (roof) operation
- Automated grain drying using fan
- Telegram alert notification
- Reduced manual effort

1.3 SYSTEM REQUIREMENTS

Hardware requirements:

- ESP32 Microcontroller
- Rain Sensor
- Soil Moisture Sensor
- DHT11 Sensor
- DC Motor
- L298D Motor Driver
- DC Fan
- DC Pump
- LCD Display (I2C)

- PIR Sensor
- Reed Switch
- LM2596 Buck Converter
- 12V DC Power Supply
- Connecting Wires

Software requirements:

- Operating System: Windows 10 / Windows 11
- Arduino IDE
- Telegram Application

1.4 PROBLEM DEFINITION:

After harvesting, grains are usually kept in open areas for drying. During this time, sudden or heavy rainfall can wet the grains and increase moisture content. This reduces grain quality and causes loss to farmers. In the existing system, farmers must manually watch the weather and cover the crops when rain occurs. This is difficult during night time or when farmers are not present. There is no automatic system to detect rain or moisture level, and no alert system to inform farmers. Because of this, crops may get damaged due to delayed action. Therefore, an automatic system is needed to detect rainfall, protect crops using a shield, reduce moisture using a drying fan, and send alerts to farmers.

To overcome these problems, an automatic and smart grain protection system is required. Such a system can use rain sensors to detect rainfall immediately and automatically activate a protective shield to cover the grains. At the same time, a drying fan can be used to reduce the moisture level and speed up the drying process. Additionally, a notification system such as Telegram alerts can inform farmers instantly about rain detection or system activation. This allows farmers to stay informed about the condition of their crops even when they are not physically present in the field. By implementing this automated solution, farmers can protect their harvested grains more effectively, reduce manual effort, and prevent crop loss caused by sudden rainfall. This system improves the efficiency of the drying process and ensures better quality agricultural products. It also supports modern smart farming practices by combining sensors, automation, and communication technologies to assist farmers in managing their crops more efficiently.

1.5 OBJECTIVE OF PROJECT

- To detect rainfall automatically using a rain sensor
- To monitor soil and grain moisture levels continuously
- To protect crops using an automatic shield during rainfall
- To reduce moisture content in grains using a drying fan
- To automate the grain drying and rain protection process
- To send real-time alerts to farmers using Telegram
- To reduce manual effort and crop damage
- To improve grain quality and farmer productivity

1.6 SCOPE OF OBJECT

The scope of this project is to provide an automatic and smart solution for grain drying and rain protection. The system is useful for farmers who dry grains in open fields and face problems due to sudden or heavy rainfall. By using sensors and an automated shield, the system protects crops from rain and helps reduce moisture using a drying fan. The Telegram alert feature allows farmers to receive updates without being physically present at the field. This project helps in saving time, reducing manual effort, and minimizing crop loss. It is suitable for small and medium-scale farming and can be further improved in the future by adding weather forecasting, mobile app control, or cloud data storage for better agricultural management. It is mainly suitable for small and medium agricultural fields, but it can also be expanded for large-scale use. In the future, additional features such as weather forecasting, mobile application control, solar power, and cloud data storage can be integrated to make the system more efficient and farmer-friendly.

This project helps farmers save time and effort because they no longer need to constantly watch the weather while drying grains. It also reduces the risk of crop damage caused by sudden rain and helps maintain the quality of agricultural produce. The system is especially useful for small and medium-scale farmers who depend on traditional open-field drying methods. Currently, the system is designed mainly for small and medium agricultural fields, but the concept can easily be expanded for large-scale farming areas. In the future, this project can be further improved by integrating advanced technologies such as weather forecasting systems, which can predict rainfall in advance and prepare the system automatically. A mobile application can also be developed to allow farmers to control and monitor the system remotely. Additionally, solar power can be used to make the system energy-efficient and suitable for rural areas with limited electricity supply. Cloud data storage can also be added to store environmental data, which can help farmers analyze weather patterns and improve farm management.

CHAPTER 2
LITERATURE SURVEY

2. LITERATURE SURVEY

[1] Kumar, S., Reddy, M., & Rao, A. (2025). An IoT Based Smart Grain Drying and Monitoring System.

Title: An IoT Based Smart Grain Drying and Monitoring System

Authors: Suresh Kumar, Mahesh Reddy, Anil Rao

Published: January 2025

Summary:

This paper discusses an IoT-based system designed to monitor grain moisture and environmental conditions during the drying process. Sensors are used to measure temperature, humidity, and moisture content, and a microcontroller controls the drying mechanism automatically. The system reduces manual effort and improves grain quality by preventing over-drying and moisture retention.

[2] Patel, R., Shah, N., & Mehta, K. (2025). Automated Rain Detection and Crop Protection System Using IoT.

Title: Automated Rain Detection and Crop Protection System Using IoT

Authors: Rakesh Patel, Nisha Shah, Kunal Mehta

Published: March 2025

Summary:

This paper presents an IoT-based rain detection system for protecting crops during sudden rainfall. A rain sensor is used to detect rain, and an automated covering mechanism is activated to protect crops. The authors highlight the importance of automation in reducing crop damage due to unexpected weather conditions.

[3] Singh, A., Verma, P., & Yadav, S. (2025). Smart Agriculture Monitoring System Using ESP32.

Title: Smart Agriculture Monitoring System Using ESP32

Authors: Amit Singh, Priya Verma, Sandeep Yadav

Published: June 2025

Summary:

This study focuses on the use of the ESP32 microcontroller for monitoring agricultural parameters such as temperature, humidity, and soil moisture. The system enables real-time monitoring and improves decision-making in farming practices. Results show improved efficiency compared to traditional method.

[4] Rao, K., Naidu, V., & Chandra, R. (2024). IoT Based Soil Moisture Monitoring and Control System.

Title: IoT Based Soil Moisture Monitoring and Control System

Authors: Kiran Rao, Venkat Naidu, Ravi Chandra

Published: February 2024

Summary:

This paper describes an IoT-based system for continuous soil moisture monitoring. When moisture exceeds a predefined threshold, control actions are taken automatically. The system helps maintain optimal moisture levels and reduces crop loss due to water imbalance.

[5] Das, P., Mishra, R., & Sen, A. (2024). Automated Crop Protection System for Rainfall Conditions.

Title: Automated Crop Protection System for Rainfall Conditions

Authors: Pranab Das, Ritu Mishra, Ananya Sen

Published: April 2024

Summary:

The authors propose an automated crop protection system that activates a protective shield during rainfall. The system uses rain sensors and motor-driven covers to prevent crop damage. The study emphasizes reduced human intervention and faster response time.

[6] Verma, T., Joshi, S., & Kulkarni, P. (2024). IoT Based Grain Storage and Drying System.

Title: IoT Based Grain Storage and Drying System **Authors:**

Tushar Verma, Shubham Joshi, Pranav Kulkarni **Published:**
July 2024.

Summary:

This paper introduces an IoT-enabled grain storage and drying system that monitors moisture and temperature to maintain grain quality. Automated fans are used for drying, and remote monitoring improves system reliability.

[7] Ahmed, S., Khan, M., & Ali, R. (2024). Smart Weather-Based Crop Protection Using IoT.

Title: Smart Weather-Based Crop Protection Using IoT

Authors: Salman Ahmed, Mohammad Khan, Rehan Ali

Published: September 2024.

Summary:

This research focuses on weather-based crop protection using IoT sensors. Rain and humidity data are analyzed to control protective mechanisms automatically. The system helps farmers protect crops from sudden weather changes.

[8] Nair, A., Menon, R., & Pillai, S. (2023). IoT Based Automated Farming System Based with Remote Alerts.

Title: IoT Based Automated Farming System with Remote Alerts

Authors: Arun Nair, Ramesh Menon, Suresh Pillai

Published: January 2023

Summary:

This paper presents an automated farming system that sends real-time alerts to users through mobile applications. The system improves monitoring efficiency and allows farmers to respond quickly to critical conditions.

[9] Gupta, V., Agarwal, S., & Jain, R. (2023). Smart Crop Monitoring and Protection Using IoT Technology.

Title: Smart Crop Monitoring and Protection Using IoT Technology

Authors: Vishal Gupta, Sneha Agarwal, Rohit Jain

Published: March 2023.

Summary:

The authors propose an IoT-based crop monitoring system that integrates moisture, rain, and temperature sensors. Automated actions are taken to protect crops during adverse conditions, reducing crop damage and labor cost.

[10] Reddy, L., Prakash, D., & Srinivas, N. (2023). IoT Based Grain Drying System with Cloud and Alert Services.

Title: IoT Based Grain Drying System with Cloud and Alert Services

Authors: Lokesh Reddy, Dinesh Prakash, N. Srinivas

Published: June 2023.

Summary:

This paper presents a grain drying system using IoT and alert services. Moisture and environmental data are monitored continuously, and alerts are sent to users during abnormal conditions. The system improves drying efficiency and prevents

2.1 LITERATURE SURVEY

S.No	Paper Title	Authors	Methodology	Advantages	Disadvantages	Future Enhancements
1	An IoT Based Smart Grain Drying and Monitoring System	Kumar et al.	Sensors check moisture and control drying automatically	Better grain quality	No cloud support	Add mobile alerts
2	Automated Rain Detection System	Patel et al.	Rain sensor detects rain and covers crops	Protects crops from rain	No weather prediction	Weather forecast
3	Smart Agriculture System Using ESP32	Singh et al.	ESP32 monitors temperature and moisture	Low cost, real-time data	No automation	Automatic control
4	IoT Based Soil Moisture System	Rao et al.	Soil sensor controls water supply	Saves water	Only irrigation	Crop protection
5	Automated Crop Protection System	Das et al.	Rain sensor activates motor cover	Quick response	No drying system	Grain drying add-on
6	IoT Based Grain Storage System	Verma et al.	Sensors and fans dry grains	Prevents spoilage	No alerts	Mobile alerts
7	Weather Based Crop Protection	Ahmed et al.	Weather sensors protect crops	Handles sudden rain	Internet needed	Offline mode
8	Automated Farming with Alerts	Nair et al.	Sensors send alerts to mobile	Easy monitoring	Mobile dependent	Voice alerts
9	Smart Crop Monitoring System	Gupta et al.	Multiple sensors monitor crops	Reduces labor	Complex setup	AI support
10	IoT Grain Drying with Cloud	Reddy et al.	Cloud-based monitoring system	Efficient drying	Needs internet	Local control

TABLE : 2.1 LITERATURE SURVEY

CHAPTER 3
SYSTEM ANALYSIS

3. SYSTEM ANALYSIS

3.1 SYSTEM ARCHITECTURE:

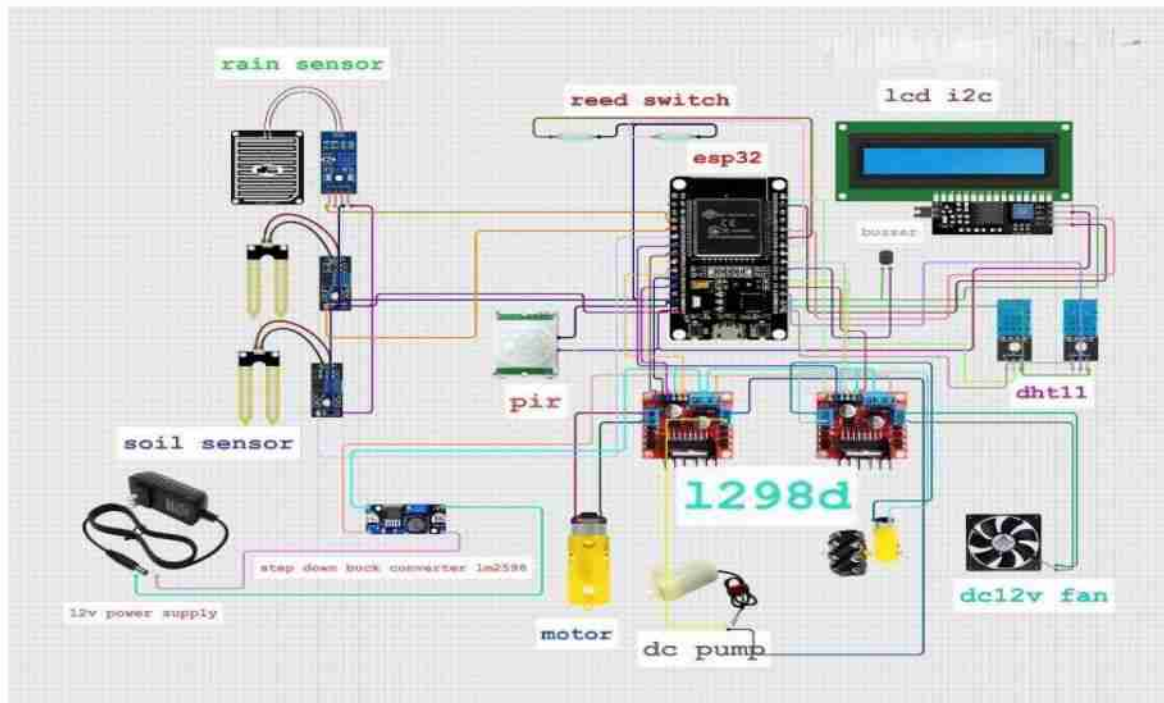


Fig 3.1 IOT BASED AUTOMATED GRAIN DRYING AND RAIN PROTECTION USING TELEGRAM ALERTS

The architecture diagram shows the overall working structure of the automated agricultural protection system. It consists of rain sensors, moisture sensors, and fire detection sensors connected to a controller unit. The controller processes the sensor data and controls the shield roof mechanism to open or close based on weather conditions. When rain is detected, the shield opens to protect crops, and during dry conditions it allows grain drying. The fire sensor continuously monitors the field or storage area and sends alerts if fire is detected. All components work together to provide automatic crop protection, grain drying, and fire safety. This layered architecture ensures separation of concerns, improved maintainability, and enhanced system performance.

3.2 ESP32 MICROCONTROLLER

The ESP32 microcontroller is a low-cost, powerful microcontroller with built-in Wi-Fi and Bluetooth features. It is used to collect data from sensors, process the information, and control devices like motors, fans, and alarms. In this project, ESP32 acts as the main control unit, managing rain detection, grain drying, and fire detection automatically.

3.3 ESP32 FEATURES

- Built-in Wi-Fi and Bluetooth support
- High processing speed with dual-core processor
- Low power consumption, suitable for IoT applications
- Supports multiple sensors and modules
- Provides GPIO, ADC, PWM, UART, SPI, and I2C interfaces
- Can be programmed using Arduino IDE
- Supports real-time data monitoring
- Compact size and cost-effective

3.4 GRAIN DRYING

Grain drying is an important agricultural process carried out after harvesting crops like rice, wheat, maize, and pulses. Freshly harvested grains contain a high amount of moisture. If this moisture is not removed, grains can get spoiled due to the growth of fungi, bacteria, and insects.

3.5 GRAIN DRYING FEATURES

- Reduces excess moisture from harvested grains
- Prevents spoilage caused by fungi, insects, and bacteria
- Improves storage life of grains
- Maintains grain quality and weight
- Reduces post-harvest losses
- Helps in safe long-term storage
- increases market value of grains